6. Zigzag Conversion

Medium | Topics | Companies

The string `"PAYPALISHIRING"` is written in a zigzag pattern on a given number of rows like this: (you may want to display this pattern in a fixed font for better legibility)

```
P   A   H   N
A P L S I I G
Y   I   R
```

And then read line by line: `"PAHNAPLSIIGYIR"`

Write the code that will take a string and make this conversion given a number of rows:

```
string convert(string s, int numRows);
```

Example 1:

```
Input: s = "PAYPALISHIRING", numRows = 3
Output: "PAHNAPLSIIGYIR"
```

Example 2:

```
Input: s = "PAYPALISHIRING", numRows = 4
Output: "PINALSIGYAHRPI"
Explanation:
P     I    N
A   L S  I G
Y A   H R
P     I
```

Example 3:

```
Input: s = "A", numRows = 1
Output: "A"
```

Constraints:

- `1 <= s.length <= 1000`
- `s` consists of English letters (lower-case and upper-case), `','` and `'.'`.
- `1 <= numRows <= 1000`

Accepted 1158 / 1158 testcases passed

Akankshyaa submitted at Nov 01, 2026 14:22

Analysis | Solution

Save 41%

Big Diwali Dhamaka

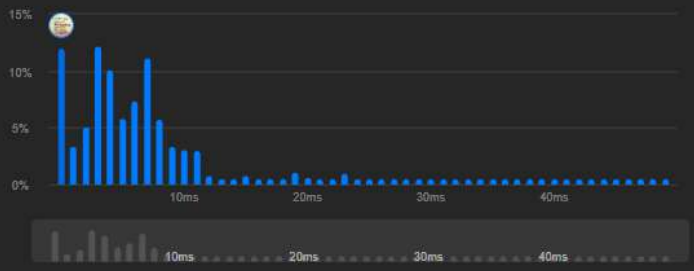
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Runtime

0 ms | Beats 100.00%

Memory

11.00 MB | Beats 91.84%



Code | C++

```cpp
class Solution {
public:
    string convert(string s, int numRows) {
        if (numRows == 1 || numRows >= s.size())
            return s;
        string result = "";
        int cycleLen = 2 * numRows - 2;
        for (int row = 0; row < numRows; row++) {
```

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```cpp
class Solution {
public:
    string convert(string s, int numRows) {
        if (numRows == 1 || numRows >= s.size())
            return s;
        string result = "";
        int cycleLen = 2 * numRows - 2;
        for (int row = 0; row < numRows; row++) {
            for (int j = 0; j + row < s.size(); j += cycleLen) {
                result += s[j + row];
                int diagonal = j + cycleLen - row;
                if (row != 0 && row != numRows - 1 &&
                    diagonal < s.size()) {
                    result += s[diagonal];
                }
            }
        }
        return result;
    }
};
```

Saved

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2 | Case 3

Input

s =

"PAYPALISHIRING"

numRows =

3

Output

"PAHNAPLSIIGYIR"

Expected

"PAHNAPLSIIGYIR"

Parameter	Row Simulation	Mathematical Pattern
Time complexity	$O(n)$	$O(n)$
Auxiliary space	$O(n)$	$O(1)$
Extra data structure	Vector of strings	None
Direction tracking	Required	Not required
Implementation	Easier	Slightly mathematical
Efficiency	Good	Better space efficiency